



Cairo University



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Advanced Tic Tac Toe Game

Performance Measurement and Optimization

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6. Performance Measurement and Optimization:

Authentication Timing	0.002607 seconds			
Metric	Ai_easy	Ai_medium	Ai_hard	Two player
Player Move Timing average(sec)	2.68751	3.0015	2.90961	3.4×10^{-3}
AI Move Timing average(sec)	1.01623	1.02231	1.02837	-----
Game Status Check Timing average for ai	1.03232	1.03538	1.04202	-----
Game Status Check Timing average for player	2.70067	3.01985	2.92207	4.05065
Game Reset Timing average(sec)	3.10511	3.52228	1.7×10^{-5}	4.07995
Game History Saving Timing(sec)	1×10^{-3}	9.7×10^{-4}	9.7×10^{-4}	1e-06
Finding Best Move Timing average	900 nsec	5.9e-05	0.023294	-----
Menu Interaction Timing	25.7237	54.8565	38.7111	65.1612

Optimization:

- Use Arrays Instead of Vectors: using arrays can be faster than vectors due to lower overhead.
- Optimizing AI Move Calculation: The AI uses the minimax algorithm with alpha-beta pruning to determine the best move.
- Code Refactoring: Separate concerns such as UI updates, game logic, and AI decision-making into different functions or classes.

- Reuse Board Initialization Constant Avoid recreating the initial board vector repeatedly.
- Avoid Reopening History File Repeatedly History filename is used multiple times.
- Reduce Repetitive Output Streams For cleaner code
- Refactor Repetitive Input Validation Extract a reusable function for input validation
- Optimizing Memory Usage:203MB
- CPU Utilization: 0.1%